Abstractive News Summarization using BART and Mistral

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Project Description:

We propose to build an abstractive news summarization model using BART and Mistral. The purpose of our project is to develop efficient text summarization models that can generate concise and informative summaries from lengthy news articles. With the huge volume of news articles published daily, users will find it difficult to read every article in a short period of time. Our project aims to address this challenge by using the latest large language models to produce accurate and meaningful summaries, preserving the essence of the original articles while eliminating redundancy.

Data Source: Kaggle Link for dataset: https://www.kaggle.com/datasets/gowrishankarp/newspaper-text-summarization-cnn-dailymail/data

This dataset contains articles and summaries of 300k+ news from CNN and Daily mail. The 'article' column has the full text of the news and 'highlight' column has the summary for the news. There are separate csv files for train, test and validation of 1.37GB news. Due to GPU constraints, we plan randomly sample 3000 records from train.csv (which is 12MB of original size 1.26GB) and we will use the subset for training and testing split.

Research paper links: BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension https://arxiv.org/pdf/1910.13461v1

Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer https://arxiv.org/pdf/1910.10683v4

Approaches:

1. Both teammates will work in data preprocessing methods. We plan to remove the extra characters, html tags using regular expressions. Tokenization must be done using BartTokenizer (which uses Byte-Pair Encoding) for BART model and AutoTokenizer (which used SentencePiece) for Mistral model.
2. Teammate 1 will work on pretrained Mistral model and fine-tuning BART model for text summarization.
3. Teammate 2 will work on pretrained BART model and fine-tuning Mistral model for text summarization.
4. We will implement traditional BART using facebook/bart-large-cnn which is Seq2seq model and traditional Mistral using mistralai/Mistral-7B which is decoder only model.
5. We plan to do fine tuning using LoRA (Low Rank Adaptation) technique. For BART, LoRA can be applied to both encoder and decoder. For Mistral, LoRA can be applied to decoder self-analysis layer.
6. The following evaluation metrics for actual and generated summaries will be used: ROUGE (Recall-Oriented Understudy for Gisting Evaluation), BERTScore for BART, Perplexity for Mistral.

Computational Resource: Google Colab and Kaggle's T4 GPU